Ch 8.2.1, 8.2.2: Bagging and Random Forests

Lecture 25 - CMSE 381

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

Fri, Mar 27, 2026

Announcements

Last time:

- 8.1 Decision Trees - regression
- 8.1 Decision Trees - classification

This lecture:

- 8.2.1 Bagging
- 8.2.2 Random forest

Announcements:

- Homework 5 Due Sunday

21	W	3/18	Polynomial & Step Functions	7.1-7.2		
22	F	3/20	Step Functions; Basis functions; Start Splines	7.2-7.4		
23	M	3/23	Regression Splines	7.4		Q7
24	W	3/25	Decision Trees	8.1		
25	F	3/27	Random Forests	8.2.1, 8.2.2	HW #5 Due Sun 3/29	
26	M	3/30	Maximal Margin Classifier	9.1		Q8
27	W	4/1	SVC	9.2		
28	F	4/3	SVM	9.3, 9.4		
29	M	4/6	Single Layer NN	10.1		Q9
30	W	4/8	Multi Layer NN	10.2		
31	F	4/10	CNN	10.3	HW #6 Due Sun 4/12	
32	M	4/13	Unsupervised learning / clustering	12.1, 12.4		Q10
33	W	4/15	Virtual: Project Office Hours			
	F	4/17	***Review***			
	M	4/20	**Midterm #3**			
	W	4/22				
	F	4/24			**Proiect Due**	

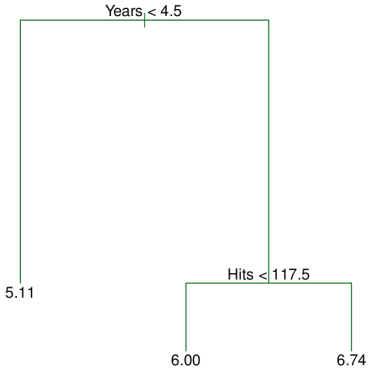
Section 1

Previously

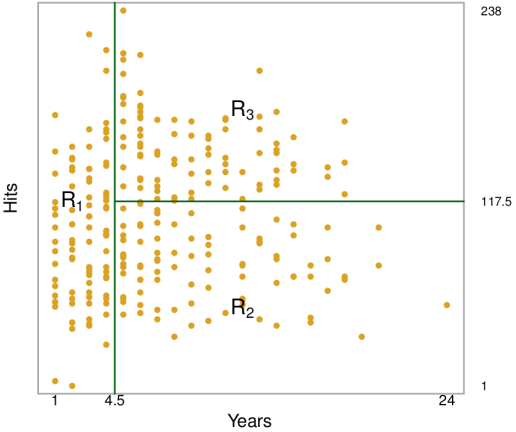
First decision tree example

	Hits	Years	LogSalary
1	81	14	6.163315
2	130	3	6.173786
3	141	11	6.214608
4	87	2	4.516339
5	169	11	6.620073
...	...	...	...
317	127	5	6.551080
318	136	12	6.774224
319	126	6	5.953243
320	144	8	6.866933
321	170	11	6.907755

- Top split assigns observations with Years$<$ 4.5to left branch
- Return mean response for players with that.
- Predictions:
  - mean log salary is 5.107, so returns $\exp(5.107) = \$165.174$ thousand dollars
  - $5.999 \Rightarrow \$402,834$
  - $6.740 \Rightarrow \$845,346$

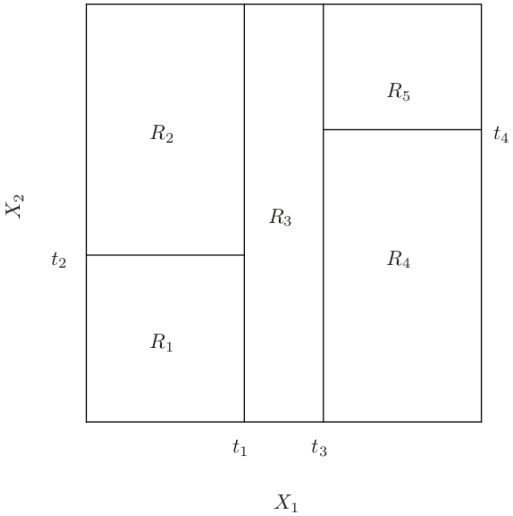


Viewing Regions Defined by Tree



How do we actually get the tree? Two steps

1. We divide the predictor space – that is, the set of possible values for $X_1, X_2, \cdots, X_p$ — into J distinct and non-overlapping regions, $R_1, R_2, \cdots, R_J$.
2. For every observation that falls into the region R_j, we make the same prediction = the mean of the response values for the training observations in R_j.



Recursive binary splitting

Goal:
Find boxes $R_1, \cdots, R_J$ that minimize

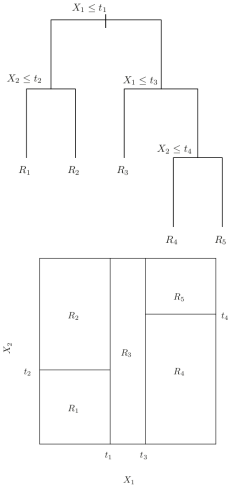
$$\sum_{j=1}^{J} \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2$$

$\hat{y}_{R_j}$ = mean response for training observations in jth box

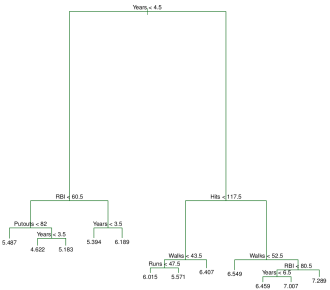
Pick s so that splitting into $\{X \mid X_j < s\}$ and $\{X \mid X_j \geq s\}$ results in largest possible reduction in RSS:

$$R_1(j,s) = \{X \mid X_j < s\}$$
$$R_2(j,s) = \{X \mid X_j \geq s\}$$

$$\sum_{i \mid x_i \in R_1(j,s)} (y_i - \hat{y}_{R_1})^2 + \sum_{i \mid x_i \in R_2(j,s)} (y_i - \hat{y}_{R_2})^2$$

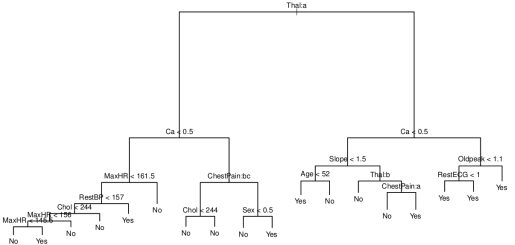


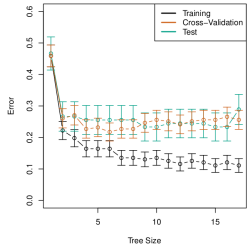
Pruning

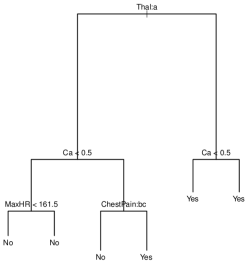


- Big trees leave you open to potential overfitting
- One option is maximum deapth
- Another is Weakest link pruning

Classification version







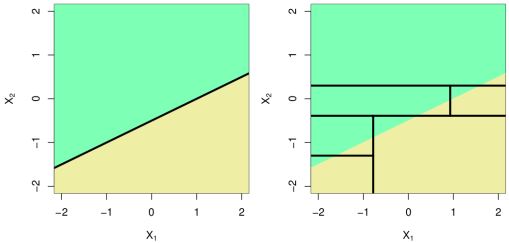
Evaluating the splits:

- $\hat{p}_{mk}$ = proportion of training observations in R_m from the kth class
- Error: $E = 1 - \max_k(\hat{p}_{mk})$
- Gini index:

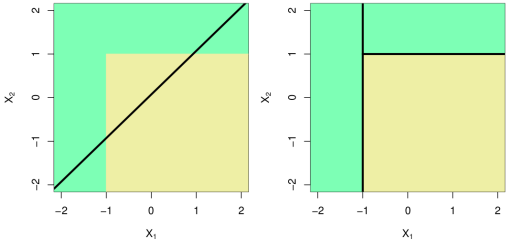
$$G = \sum_{k=1}^{K} \hat{p}_{mk}(1 - \hat{p}_{mk})$$

Linear models vs trees

- Trying to predict green vs yellow



Obviously linear does better here



Not going to beat this case though

What will your learn today?

- What does bagging of decision trees accomplish?
- How do you use out of bag error estimation for decision trees?
  - You should be able to describe this for both regression and classification trees.
- What problem of bagging does using random forest address?
- What is the relationship between random forest and bagging?

Section 2

8.2.1 Bagging

Use ensemble of trees to reduce variance

Want to do (but can't):
Build separate models from independent training sets, and average resulting predictions:

- $\hat{f}^1(x), \cdots, \hat{f}^B(x)$ for B separate training sets
- Return the average

$$\hat{f}_{avg}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^b(x)$$

- Decreases variance, but not practical

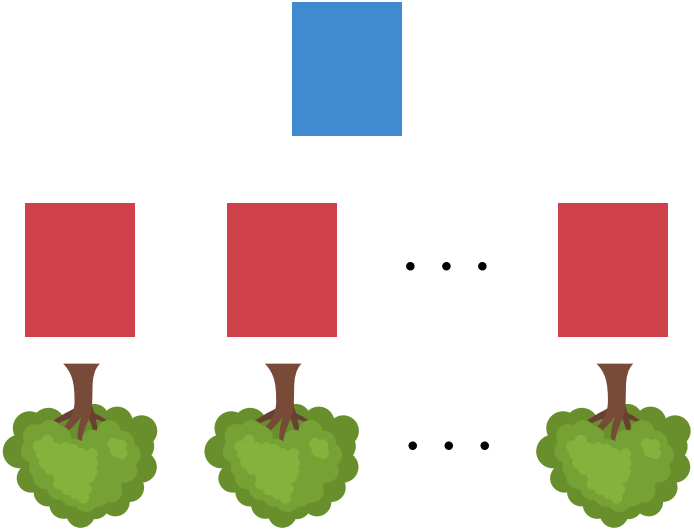
Boostrap modification:

- Work with fixed data set
- Take B samples from this data set (with replacement)
- Train method on bth sample to get $\hat{f}^{*b}(x)$
- Return average of predictions (regression)

$$\hat{f}_{bag}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^{*b}(x)$$

or majority vote (classification)

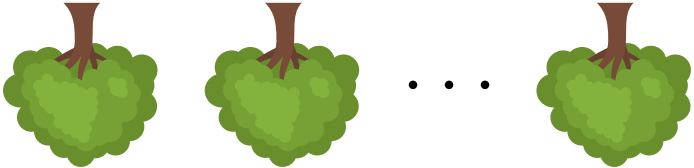
Tree version



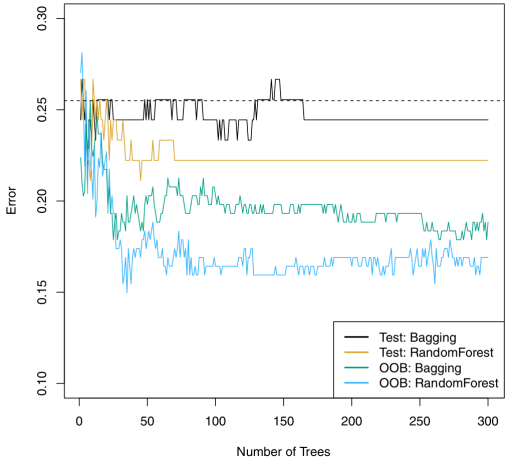
- *Blue is original data set* $X : n \times p$
- *Red are the subsampled data sets*
- *Build a tree from each*
- *Get prediction from each, return the average*

Prediction on new data point

New data point goes in, run through each tree, average outputs



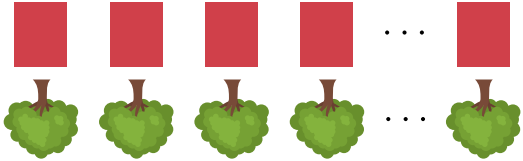
Example: Heart classification data



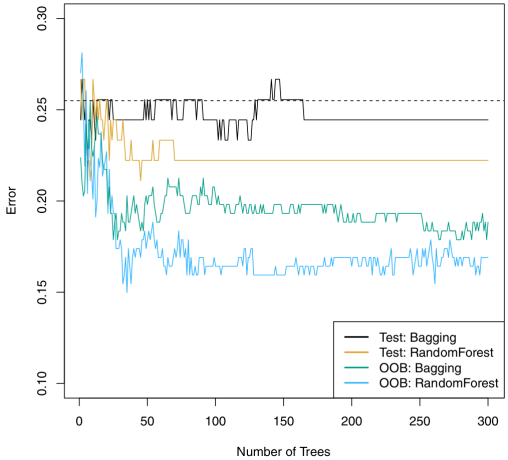
- Example is done for classification
- Ignore everything but black lines
- Dashed is test error from single classification tree
- Solid black is from bagged version
- Note that you just need B big enough for error to stabilize, so about 100 is good
- emphasize that number of trees is not a flexibility parameter

Out of Bag Error Estimation

- On average, bootstrap sample uses about 2/3 of the data *We did this homework problem a while ago I think? Ch 5, Ex. 2*
- Remaining observations not used are called *out-of-bag* (OOB) observations
- For each observation, run through all the trees where it wasn't used for building
- Return the average (or majority vote) of those as test prediction! rather law of large number
- Bootstraped version of LOOCV.



Error using OOB



- Green line has OOB error
- note OOB is averaged B/3 predictions per observation
- Book says the fact that the OOB value is so much lower is just random chance?

Test your understanding: PollEv

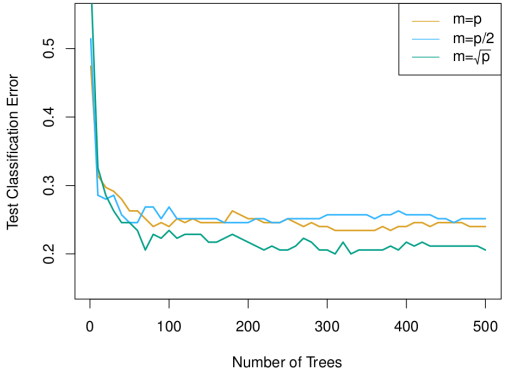
Section 3

Random Forests

The idea

- Goal is to decorrelate the bagged trees:
  - If there is a strong predictor, the first split of most trees will be the same
  - Most or all trees will be highly correlated
  - Averaging highly correlated quantities doesn't decrease variance as much as uncorrelated
- The random forest fix:
  - Each time a split is considered, only use a random subset of m the predictors
  - Fresh sample taken every time
  - Typically $m \approx \sqrt{p}$
  - On average, $(p - m)/p$ of splits won't consider strong predictor
  - $m = p$ gives back bagging

Example on gene expression



- Helpful for large p
- Predict one of 14 types of cancer (15 total levels), use 500 genes with most variance
- Random split to train and test set, three values of m
- Error rate of single tree: 45.7%
- Null rate 75.4% (classify each into the dominant overall class, here the healthy class)
- 400 trees enough for good performance
- $m = \sqrt{p}$ small improvement over bagging ($m = p$) in this case

Coding time!

TL:DR

- Bagging: trees grown independently on random samples. Trees tend to be similar to each other, can result in getting caught in local optima
- Random forest: trees independently on samples, but split is done using random subset of features

Next time

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